



Stability prediction method of time-varying real-time hybrid testing system on vehicle-bridge coupled system

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ABSTRACT

In recent years, real-time hybrid testing (RHTT) has been applied for the dynamic testing of high-speed trains running on bridges. A guarantee of stability for the RHTT system is essential to achieve a safe and reliable result. However, the inherent time-varying characteristics of the vehicle-bridge coupled system pose challenges to RHTT stability prediction. This study aims to develop a stability prediction method specifically tailored for time-varying RHTT system. Firstly, the vehicle-bridge coupled RHTT was modelled using a discrete state-space representation with a comprehensive consideration of the time-varying vehicle-bridge interaction and the dynamics of the shaking table. Subsequently, a time-varying stability criterion was derived from the periodic time-varying state matrix, forming the basis for a relative stability prediction method. The validity of the proposed method was confirmed through simulations and experiments employing a single-axle interaction within the vehicle-bridge coupled RHTT system. The coupled system consisted of a quarter-car model and simply supported beams were used as an example to evaluate the stability and accuracy of the time-varying RHTT. The results showed that the stability increased with increasing vehicle speed. Reducing the pure time delay of the shaking table improved both stability and accuracy. Increasing the effective frequency of the shaking table improved the accuracy but may reduce the stability of the RHTT.

1. Introduction

High-speed railways have developed rapidly in recent years. By the end of 2022, the total global high-speed railway mileage had reached 67700 km, with China accounting for a substantial 40267 km. Within China's high-speed railway infrastructure, approximately 57.7 % constitutes railway bridges, primarily consisting of prestressed, simply supported concrete box beams [1,2]. Common dynamic testing methods for train-bridge coupled systems include bench testing [3], field testing [4], and scaled-model testing [5,6]. However, owing to the high speeds of trains, certain limitations arise when using these methods to test train-bridge coupling vibrations. For example, the bench testing fails to simulate the interaction between a moving vehicle and a bridge, while the field tests tend to be financially burdensome. Scaled-model testing faces difficulty simulating high-speed running conditions constrained by the size limitations of the bridge specimens.

Real-time hybrid testing (RHTT) has emerged as a valuable tool for the dynamic testing of high-speed trains running on bridges

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[7,8]. The basic concept of real-time hybrid test is to split the overall system into two parts, where the hard-to-model parts are considered physical substructures and tested using actuators or shaking tables, and the easy-to-model parts are modeled and solved numerically as the numerical substructures [9,10]. RTHT has been widely used to study seismic isolation [11], structural vibration control [12,13], soil-structure interactions [14], and several other engineering problems. The technical complexity of real-time hybrid testing lies in the simultaneous completion of high-speed solutions of numerical substructures and precise loading on the physical substructure. Various time-integration algorithms [15–17] and control strategies [18–20] were developed to address these difficulties. Yet, unavoidable errors remain in both numerical solutions and physical loading of the RTHT, potentially reducing test result accuracy and, in some cases, leading to instability. Therefore, studying the stability of RTHTs remains critical to ensure their feasibility and safety.

Numerous studies have proposed diverse methodologies to analyze the stability of real-time hybrid testing (RTHT). Horiuchi et al. [21] regarded the loading error as a pure time delay and employed the concept of negative damping to describe the effect of this time delay on RTHT stability. Instability occurs when negative damping caused by the time delay exceeds that of the original system. Wallace et al. [22] derived an exact solution for the critical delay in stability. Gao et al. [23] extended the negative damping theory to analyze RTHT with multiple degrees of freedom. In addition to negative damping theory, Chi et al. [24] and Zhu et al. [25] utilized the discrete root locus method, employing Padé approximations to transform time delays into transfer function and assess the influence of time-integration algorithms on RTHT stability. Hong et al. [26] adopted a transfer function to represent the dynamics of the loading system, developing a stability analysis method based on gain margin principles, which jointly considers physical test system continuity and numerical substructure discreteness. While these studies usually modeled RTHT as linear system, some delve into nonlinear RTHT stability. Enokida et al. [27,28] utilized the Nyquist criterion to develop a stability prediction method for RTHT with linear or nonlinear substructure control and discussed the stability of substructures with bilinear or trilinear hysteresis. Wu et al. [29] and Zhou et al. [30,31] modelled RTHT with bilinear stiffness based on discrete transfer functions, determining the stability of the nonlinear RTHT based on pole distribution. Huang et al. [32,33] equated the nonlinear error of the loading system to time-varying delay, proposing a sufficient condition for a stable RTHT via Lyapunov-Krasovskii function.

The systems studied in existing RTHT stability research have traditionally been simple models like single-degree-of-freedom or basic multi-degree-of-freedom systems. Stability prediction methods in these studies are tailored exclusively to linear and time-invariant RTHTs. However, the stability prediction in the vehicle-bridge coupled RTHT systems introduces various complexities, including time variability (stemming from the relative motion between the vehicle and bridge), time asynchrony (owing to the integration step of the complex bridge model exceeding the loading system's interval), and multi-input-multi-output interactions (resulting from multi-interface interactions between the train and bridge). These issues present significant new challenges for RTHT stability assessment. This paper primarily focuses on addressing the challenges associated with stability prediction arising from time variability. This study simplified the coupled system to a single-axis interaction system, comprising a quarter-car model and simply supported beams. Subsequently, a novel stability prediction method is proposed to address the time-varying interaction challenge. To explain this method, Section 2 introduces the dynamic equations of each subsystem and derives a discrete state-space representation of the overall RTHT. Based on the discrete state-space representation, Section 3 establishes a stability criterion for the time-varying RTHT and develops a method for predicting relative stability. Section 4 presents practical vehicle-bridge coupled RTHTs to verify the proposed method, with detailed test information and results. Section 5 presents a parametric study assessing the stability and accuracy of

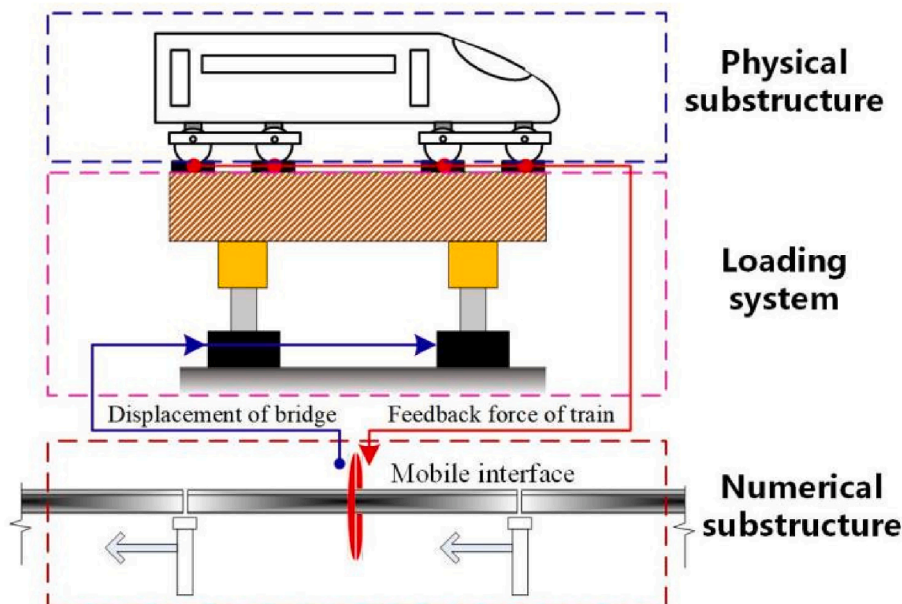


Fig. 1. Schematic of real-time hybrid test on vehicle-bridge coupled system.

the time-varying RHTT. The conclusions of this study are summarized in Section 6.

2. Dynamic model of the vehicle-bridge coupled RHTT

The vehicle-bridge coupled system can be considered as two subsystems: the vehicle subsystem and bridge subsystem. In this RHTT setup, a relative coordinate system is adopted, placing the vehicle as the horizontally stationary origin while the bridge moves backward to simulate the relative motion. This enables the vehicle subsystem to act as the physical substructure for physical testing, while the bridge as the numerical substructure for real-time calculations. The vertical displacement of the bridge at the wheel-rail contact point was loaded onto the vehicle via a shaking table. Subsequently, the vehicle reaction force was measured by the acquisition system and then fed back into the numerical substructure. The overall framework of the real-time hybrid test system is shown in Fig. 1.

The stability analysis of the RHTT in existing studies is typically based on the transfer function [23–31]. However, due to the time-varying characteristics of the vehicle-bridge coupled system, deriving the RHTT transfer function becomes challenging. Furthermore, considering the discrete integration algorithm used for the numerical substructure, this study models the vehicle-bridge coupled RHTT based on a discrete state-space representation.

2.1. Numerical substructure

Railway bridges commonly utilize prestressed, simply supported concrete box beams [1,2]. In this study, the bridge was modelled as a structure composed of a series of simply supported beams. Taking a simply supported beam as an example, a schematic of the beam subjected to a moving load is shown in Fig. 2.

In Fig. 2, $u(x, t)$ represents the vertical displacement of the bridge at position x and time t . $P(t)$ denotes the moving load on the bridge, encompassing both the gravitational force exerted by the vehicle and the reaction force of the vehicle resulting from the vibration and irregularities of the bridge. This load moves horizontally along the bridge at speed v . L is the length of the single-span bridge, and $\delta(x)$ is the Dirac delta function. The differential equation for the bridge vibration is as follows:

$$m \frac{\partial^2 u(x, t)}{\partial t^2} + c \frac{\partial u(x, t)}{\partial t} + EI \frac{\partial^4 u(x, t)}{\partial x^4} = \delta(x - vt) P(t) \quad (1)$$

where m denotes the mass per unit length of the bridge, EI is the flexural stiffness of the bridge, and c is the damping coefficient. The modal dynamic equilibrium equation of the bridge is obtained based on the characteristics of the Dirac function and modal analysis method:

$$\mathbf{M}_q \ddot{\mathbf{q}}(t) + \mathbf{C}_q \dot{\mathbf{q}}(t) + \mathbf{K}_q \mathbf{q}(t) = \boldsymbol{\phi}(vt) P(t) \quad (2)$$

where $\mathbf{M}_q$, $\mathbf{C}_q$, and $\mathbf{K}_q$ are the modal mass, damping, and stiffness matrices of the bridge, respectively. $\mathbf{q}$ is the modal displacement vector and $\boldsymbol{\phi}$ is the mode shape vector. The expressions for these parameters are presented in Appendix I.

Various time integration algorithms have been developed to solve numerical substructures [15–17]. The solving processes of these algorithms can be expressed as discrete state-space representations. Considering the typical Newmark- β method [15] as an example, the state equation for solving the bridge response is shown in Eq. (3):

$$\mathbf{z}_{b,i+1} = \mathbf{A}_b \mathbf{z}_{b,i} + \mathbf{B}_{b,i+1} P_{i+1} \quad (3)$$

where $\mathbf{z}_{b,i}$ is the state vector of the bridge at the i -th moment and includes the current modal displacement, velocity, and acceleration. $\mathbf{A}_b$ is the state matrix of the numerical substructure; $\mathbf{B}_{b,i}$ is the input matrix; and P_i is the moving force on the bridge at the i -th moment. The expressions for $\mathbf{A}_b$, $\mathbf{B}_{b,i}$ and $\mathbf{z}_{b,i}$ are shown in Appendix I.

Eq. (3) offers a solution specifically tailored to a single simply supported beam. However, in the vehicle-bridge coupled RHTT system, the bridge model comprises multiple simply supported beams. Taking advantage of the decoupled vibrations among these beams, it becomes necessary to solve only for the response of the beam currently interacting with the vehicle, while the remaining

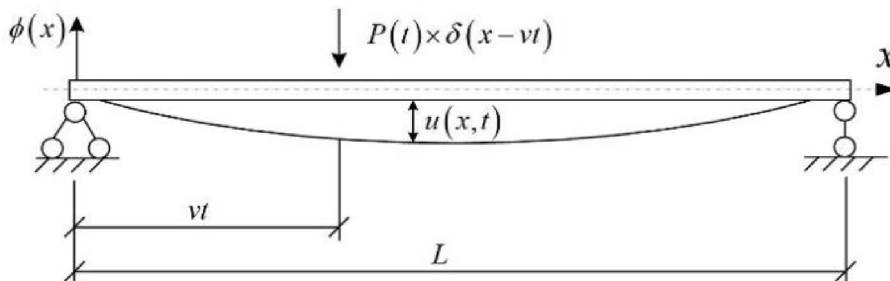


Fig. 2. Simply supported beam under moving force.

uncontacted beams remain static. This distinct characteristic enables the simplification of multi-span bridges into equivalent models resembling single simply supported beam. The scenario of a vehicle running on multi-span bridges can be treated as repeated runs on single-span beam. As the vehicle progresses from one bridge span to the next, the states of the simply supported beam are reset to zero. This reset mirrors the vehicle entering a new stationary span, simulating realistic conditions. The schematic representation of this equivalent analysis model, based on a single simply supported beam, is depicted in Fig. 3.

The numerical representation of the equivalent analysis model can be obtained by modifying the state-space representation of the simply supported beam, in which only the state matrix needs to be modified:

$$\begin{aligned} \mathbf{z}_{b,i+1} &= \mathbf{A}_{b,i} \mathbf{z}_{b,i} + \mathbf{B}_{b,i+1} P_{i+1}, \\ \mathbf{A}_{b,i} &= \begin{cases} \mathbf{0}_{n_b \times n_b} & i = mN + 1, m \in \mathbf{N}^+ \\ \mathbf{A}_b & i \neq mN + 1, m \in \mathbf{N}^+ \end{cases}, \end{aligned} \quad (4)$$

where $\mathbf{0}$ is a matrix with all elements equal to 0, and the subscripts represent the number of rows and columns in the sequence. n_b stands for the dimensions of the state vector of the numerical substructure. m represents the m -th span of the bridge, and N represents the vehicle with N contact nodes for each single-span bridge. $\mathbf{N}^+$ represents the set of positive integers.

The dynamic response of the bridge can be obtained by superimposing the modal responses. In the vehicle-bridge coupled RHTT, the numerical substructure only needs to output the vertical response x_I at the wheel-rail contact point, as shown in Eq. (5):

$$x_I(t) = u(vt, t) = \sum_{i=1}^{n_q} (q_i \times \phi_i(vt)) \quad (5)$$

where n_q indicates that the numerical substructure considers the first n_q -order mode of the simply supported beam. The output equation of the numerical substructure is derived from Eq. (5) as follows:

$$x_{I,i+1} = \mathbf{C}_{b,i+1} \mathbf{z}_{b,i+1} \quad (6)$$

where $\mathbf{C}_{b,i}$ represents the output matrix of the bridge, and its expression is given in Appendix I.

The state equation (Eq. (4)) and output equation (Eq. (6)) collectively constitute the state-space representation of the numerical substructure, accounting for the single-axis interaction between the vehicle and bridge. As for the case of a convey of vehicles, with the adoption of an appropriate equivalent analysis model for the bridge, the scenario of train operation on multi-span bridges can similarly be perceived as a repetitive operation on the equivalent model.

2.2. Physical test system

Existing research has indicated that the performance of a shaking table is affected by the dynamics of physical specimens [27]. To simplify the interaction analysis, this study modelled the shaking table and specimen as a unified system, named the physical test system, where the input is the displacement command x_{Input} of the shaking table and output is the reaction force F_f of the physical specimen (Fig. 4).

In terms of the physical substructure, various vehicle models were developed, such as the moving mass [34], quarter-car model [35], and multibody dynamics model [36]. The motion of the aforementioned vehicle models can be described using dynamic equilibrium equations. This paper takes the quarter-car model as an example to represent the physical substructure, as shown in Fig. 4. Here, m_c represents the mass of the carriage, m_b stands for the mass of the bogie, and m_w denotes for the mass of the wheelset. k_c and c_c denote the stiffness and damping coefficients of the secondary suspension, respectively. k_b and c_b represent the stiffness and damping coefficients of the primary suspension, respectively. x_c , x_b , and x_w correspond to the vertical displacements of the carriage, bogie, and

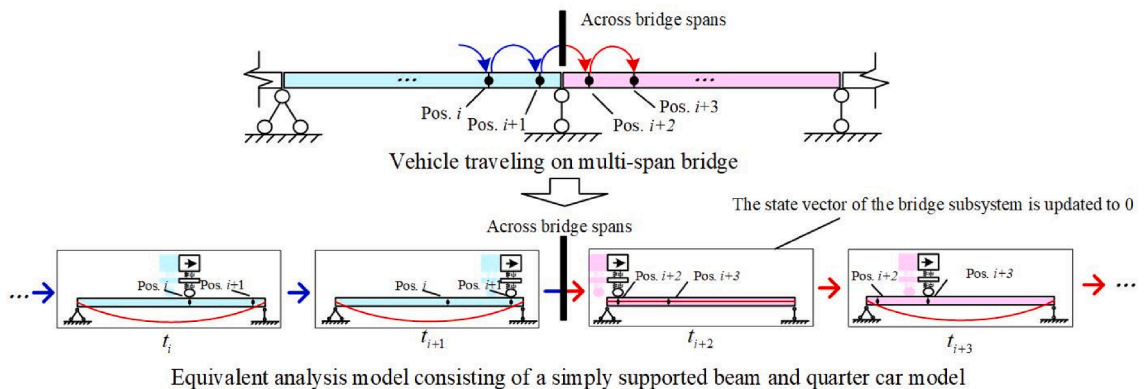


Fig. 3. Equivalent analysis model of multi-span bridge based on a single simply supported beam.

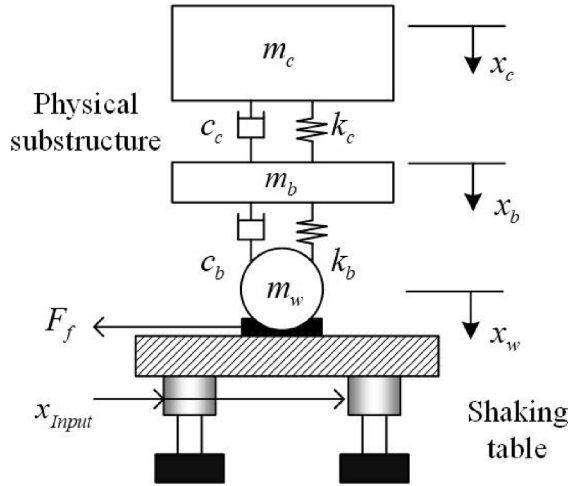


Fig. 4. Physical test system.

wheelset, respectively, relative to the railway. The wheel-rail interaction was simplified as a vertical tight contact [37], hence x_w is always equal to x_g . The dynamic equilibrium equation is given by Eq. (7):

$$\mathbf{M}_v \ddot{\mathbf{x}}_v(t) + \mathbf{C}_v \dot{\mathbf{x}}_v(t) + \mathbf{K}_v \mathbf{x}_v(t) = -\mathbf{M}_v \mathbf{I}_v \ddot{x}_g(t) \quad (7)$$

where $\ddot{x}_g$ is the acceleration of the shaking table; $\mathbf{M}_v$, $\mathbf{C}_v$, and $\mathbf{K}_v$ are the mass, damping, and stiffness matrices of vehicle, respectively; $\mathbf{x}_v = \{x_b, x_c\}^T$ is the displacement vector; and $\mathbf{I}_v = \{1, 1\}^T$ is the load position vector.

The reaction force of the vehicle can be calculated using Eq. (8):

$$F_f(t) = -\mathbf{I}_v^T \mathbf{M}_v \left(\mathbf{I}_v \ddot{x}_g(t) + \ddot{\mathbf{x}}_v(t) \right) - m_w \ddot{x}_g(t) \quad (8)$$

By performing the Laplace transformation on Eqs. (7) and (8), the transfer function between x_g and F_f can be obtained as

$$G_v(s) = \frac{F_f(s)}{x_g(s)} = \left(-\mathbf{I}_v^T \mathbf{M}_v (\mathbf{M}_v s^2 + \mathbf{C}_v s + \mathbf{K}_v)^{-1} (\mathbf{C}_v s + \mathbf{K}_v) \mathbf{I}_v - m_w \right) s^2 \quad (9)$$

After considering the interaction between the specimen and shaking table, the shaking table can be modelled as a transfer function G_{st} (owing to the closed-loop actuator and dynamic influence of the physical specimen) with a pure time delay τ (owing to discrete time elements in the controller and filters) [27], as shown in Eq (10):

$$\frac{x_g(s)}{x_{input}(s)} = e^{-\tau s} G_{st}(s) \quad (10)$$

Combining Eqs. (9) and (10), the transfer function between input x_{input} and output F_f of the physical testing system is obtained as follows:

$$\frac{F_f(s)}{x_{input}(s)} = e^{-\tau s} G_{st}(s) G_v(s) \quad (11)$$

The physical testing system is continuous. To obtain the discrete state-space equation of the overall RTHT, a numerical integration algorithm is needed to solve Eq. (11). Firstly, x_{g0} is defined as an intermediate variable between F_f and x_{input} to divide Eq. (11) into the transfer function and pure time delay terms, as shown in Eq. (12).

$$\frac{x_{g0}(s)}{x_{input}(s)} = e^{-\tau s} \quad (12.a)$$

$$\frac{F_f(s)}{x_{g0}(s)} = G_{st}(s) G_v(s) \quad (12.b)$$

The pure time delay is assumed to be an integer multiple of the integration time step, and the time delay terms in Eq. (12.a) can be expressed using the state-space representation, as shown in Eq. (13):

$$\begin{cases} \mathbf{z}_{d,i+1} = \mathbf{A}_d \mathbf{z}_{d,i} + \mathbf{B}_d x_{input,i+1} \\ x_{g0,i+1} = \mathbf{C}_d \mathbf{z}_{d,i+1} \end{cases} \quad (13)$$

where $\mathbf{z}_{d,i}$ is the state vector of the pure delay term in physical test system. $\mathbf{A}_d$, $\mathbf{B}_d$ and $\mathbf{C}_d$ respectively represent the relevant state matrix, input matrix and output matrix.

A discrete integration algorithm was used to solve the transfer function term in Eq. (12.b). The solution process can be organized into discrete state-space equations. Taking the fourth-order Runge-Kutta method, which is a basic algorithm in MATLAB, as an example, the state equation can be obtained as:

$$\begin{cases} \mathbf{z}_{c,i+1} = \mathbf{A}_c \mathbf{z}_{c,i} + \mathbf{B}_c \mathbf{x}_{g0,i} \\ F_{f,i+1} = \mathbf{C}_c \mathbf{z}_{c,i+1} + \mathbf{D}_c \mathbf{x}_{g0,i+1} \end{cases} \quad (14)$$

where $\mathbf{z}_{c,i}$ is the state vector of the transfer function term in physical test system. $\mathbf{A}_c$, $\mathbf{B}_c$ and $\mathbf{C}_c$ respectively represent the relevant state matrix, input matrix and output matrix. See Appendix II for the derivation and expressions of the matrices and vectors in Eqs. (13) and (14), respectively.

2.3. Real-time hybrid test on the vehicle-bridge coupled system

The real-time hybrid test was a closed-loop system composed of the aforementioned numerical, physical, and loading subsystems, as shown in Fig. 5. The external inputs to the vehicle-bridge coupled system encompassed the gravitational force exerted by the moving vehicle and the track irregularities, with the latter being considered a displacement disturbance affecting the vehicle. In the RTHT, the input x_{Input} of the shaking table comprises the numerical substructure displacement x_I and track irregularity displacement x_s . Subsequently, the output acceleration of the shaking table served as the base excitation of the physical substructure. The measured reaction force F_f of the physical substructure was then fed back to the numerical substructure, together with the vehicle gravity F_g to serve as the excitation of the numerical substructure.

As depicted in Fig. 5, the input and output matrices of the bridge demonstrate time-varying characteristics. Consequently, deriving the transfer function of the closed-loop system requires convolution operations, which makes it difficult to obtain an analytical solution. Thus, transfer-function-based stability-analysis methods are not applicable to time-varying RTHT. Therefore, this study models the real-time hybrid test based on the discrete state equation, which can be obtained by combining the state-space representations of the numerical substructure and physical test system, as shown in Eq. (15):

$$\begin{cases} \mathbf{z}_{b,i+1} = \mathbf{A}_b \mathbf{z}_{b,i} + \mathbf{B}_{b,i+1} (F_{f,i+1} + F_g) \\ x_{I,i+1} = \mathbf{C}_{b,i+1} \mathbf{z}_{b,i+1} \\ \mathbf{z}_{d,i+1} = \mathbf{A}_d \mathbf{z}_{d,i} + \mathbf{B}_d (x_{Input,i+1} + x_{s,i+1}) \\ \ddot{x}_{g0,i+1} = \mathbf{C}_d \mathbf{z}_{d,i+1} \\ \mathbf{z}_{c,i+1} = \mathbf{A}_c \mathbf{z}_{c,i} + \mathbf{B}_c \ddot{x}_{g0,i} \\ F_{f,i+1} = \mathbf{C}_c \mathbf{z}_{c,i+1} + \mathbf{D}_c \ddot{x}_{g0,i+1} \end{cases} \Rightarrow \mathbf{Z}_{i+1} = \mathbf{A}_{i+1} \mathbf{Z}_i + \mathbf{B}_{i+1} \mathbf{Y}_{i+1} \quad (15)$$

where $\mathbf{Z}_i$ is the state vector of the RTHT system at the i -th moment. $\mathbf{A}_i$ and $\mathbf{B}_i$ respectively represent the state matrix and input matrix. $\mathbf{Y}$ is the input to the coupled system, including F_g and x_s . The expressions for these parameters are provided in Appendix III. Eq. (15) is the discrete state-space representation of the vehicle-bridge coupled RTHT, capable of simulating a vehicle travelling on a multi-span bridge. Eq. (15) reveals that the state matrix of the system depends on time rather than the system's state. Consequently, the vehicle-bridge coupled system operates as a linear time-varying system, not a nonlinear system.

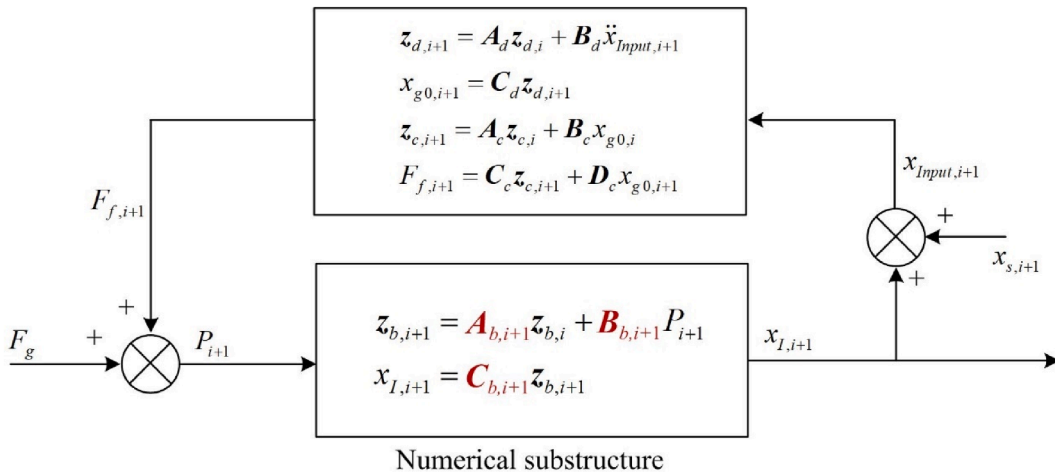


Fig. 5. Closed-loop block diagram of real-time hybrid test on vehicle-bridge coupled system.

3. Stability prediction method of the time-varying real-time hybrid test

For general linear time-invariant systems, the stability can be determined directly from the spectral radius of a discrete state matrix [38]. The system is unstable if the spectral radius is greater than 1 and stable when the spectral radius is less than 1. However, Eq. (15) illustrates that the state matrix of the time-varying RTHT evolves with time. This indicating that the stability of the system cannot be determined directly using conventional time-invariant criterion. Hence, based on the discrete state-space representation deduced in Section 2, this section introduces a stability criterion specifically tailored for the time-varying RTHT, along with a relative stability prediction method.

3.1. Stability criterion of the time-varying RTHT

The discrete state matrix signifies the error transfer relationship between consecutive states. This implies that if the spectral radius of the state matrix remains consistently below 1 throughout all time points, the absence of error amplification at any stage ensures system stability. Hence, a sufficient condition for the stability of the time-varying system can be deduced as follows:

$$\max(\rho(\mathbf{A}_i)) < 1, i \in \mathbf{N}^+ \quad (16)$$

To comprehend the stability mechanisms inherent in time-varying systems and precisely assess the stability margin of RTHT system, it becomes crucial to precisely forecast the system stability. Thus, the following demonstration aims to establish both the necessary and sufficient condition for ensuring stability of the time-varying vehicle-bridge coupled RTHT system.

Based on the discrete state-space representation shown in Eq. (15), the transfer equation between the state vectors at two consecutive time steps without excitation can be obtained as follows:

$$\mathbf{Z}_{i+x} = (\mathbf{A}_{i+x} \times \mathbf{A}_{i+x-1} \times \cdots \mathbf{A}_{i+x}) \times \mathbf{Z}_i = \prod_{j=0}^{x-1} \mathbf{A}_{i+x-j} \times \mathbf{Z}_i, x \in \mathbf{N}^+ \quad (17)$$

Based on Eq. (17), the transfer equation of the system state vectors between any two adjacent bridge spans can be further derived:

$$\begin{cases} \mathbf{Z}_{mN+1} = \prod_{j=0}^{N-1} \mathbf{A}_{mN+1-j} \mathbf{Z}_{(m-1)N+1} \\ \mathbf{Z}_{mN+2} = \prod_{j=0}^{N-1} \mathbf{A}_{mN+2-j} \mathbf{Z}_{(m-1)N+2} \\ \vdots \\ \mathbf{Z}_{(m+1)N} = \prod_{j=0}^{N-1} \mathbf{A}_{(m+1)N-j} \mathbf{Z}_{mN} \end{cases} \quad (18)$$

where $\mathbf{Z}_{mN+i}$ refers to the state vector of RTHT when the vehicle travels at the i -th contact node of the m -th bridge span.

Eq. (18) can be organized into a state transition equation:

$$\begin{Bmatrix} \mathbf{Z}_{mN+1} \\ \mathbf{Z}_{mN+2} \\ \vdots \\ \mathbf{Z}_{(m+1)N} \end{Bmatrix} = \begin{bmatrix} \prod_{j=0}^{N-1} \mathbf{A}_{mN+1-j} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \prod_{j=0}^{N-1} \mathbf{A}_{mN+2-j} & & \mathbf{0} \\ \vdots & & \ddots & \\ \mathbf{0} & \mathbf{0} & & \prod_{j=0}^{N-1} \mathbf{A}_{(m+1)N-j} \end{bmatrix} \begin{Bmatrix} \mathbf{Z}_{(m-1)N+1} \\ \mathbf{Z}_{(m-1)N+2} \\ \vdots \\ \mathbf{Z}_{mN} \end{Bmatrix} \quad (19)$$

In addition, it can be concluded from Section 2.4 that the time variability of the system originates from the time-varying matrices of the bridge subsystem. This can also be observed in the state matrix of the overall RTHT, as detailed in Appendix III. In high-speed rail systems, the majority of bridges follow fixed specifications, commonly employing simply supported beam configurations. Additionally, trains typically maintain a constant speed during their operations. In this context, the time-varying matrices of the bridge subsystem depend only on the periodically changing relative position between the vehicle and the equivalent analysis bridge model. Therefore, the state matrices of the overall RTHT change periodically, as shown in Eq. (20):

$$\mathbf{A}_i = \mathbf{A}_{i+N}, i \in \mathbf{N}^+ \quad (20)$$

Based on the periodicity shown in Eq. (20), it can be found that the state transition matrix in Eq. (19) is constant, regardless of the number of spans, as shown in Eq. (21):

$$\begin{Bmatrix} \mathbf{Z}_{mN+1} \\ \mathbf{Z}_{mN+2} \\ \vdots \\ \mathbf{Z}_{(m+1)N} \end{Bmatrix} = \begin{bmatrix} \prod_{j=0}^{N-1} \mathbf{A}_{N+1-j} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \prod_{j=0}^{N-1} \mathbf{A}_{N+2-j} & & \mathbf{0} \\ \vdots & & \ddots & \\ \mathbf{0} & \mathbf{0} & & \prod_{j=0}^{N-1} \mathbf{A}_{2N-j} \end{bmatrix} \begin{Bmatrix} \mathbf{Z}_{(m-1)N+1} \\ \mathbf{Z}_{(m-1)N+2} \\ \vdots \\ \mathbf{Z}_{mN} \end{Bmatrix} \quad (21)$$

Theoretically, it can be concluded that when the spectral radius of the state matrix in Eq. (21) is greater than 1, the vehicle-bridge coupled RHT is unstable. However, this calculation is computationally complex and can be further simplified. According to the properties of eigenvalues, that is, $\text{eig}(\mathbf{A} \times \mathbf{B}) = \text{eig}(\mathbf{B} \times \mathbf{A})$, the eigenvalues of each matrix along the diagonal of the state matrix in Eq. (21) are identical, as shown in Eq. (22):

$$\text{eig}\left(\prod_{j=0}^{N-1} \mathbf{A}_{N-j}\right) = \text{eig}\left(\prod_{j=0}^{N-1} \mathbf{A}_{N+i-j}\right), \quad i \in [1, 2, \dots, N] \quad (22)$$

Therefore, the final stability criterion for the time-varying RHT can be obtained as Eq. (23):

$$\begin{cases} \rho\left(\prod_{j=0}^{N-1} \mathbf{A}_{N-j}\right) \geq 1 & \text{Unstable} \\ \rho\left(\prod_{j=0}^{N-1} \mathbf{A}_{N-j}\right) < 1 & \text{stable} \end{cases} \quad (23)$$

where $\rho(\mathbf{A})$ stands for the spectral radius of the matrix $\mathbf{A}$, which is the maximum modulus of the eigenvalues.

3.2. Case study of the vehicle-bridge coupled RHT

To verify the accuracy of the stability criterion proposed in Eq. (24), a series of simulations were conducted based on Eq. (15). A quarter-car model of the CRH380A high-speed train was used in the simulations [8]. Detailed parameters of the carriage and bogie are listed in Table 1. The parameters of the bridge were $m = 22,691 \text{ kg/m}$, $EI = 5.2 \times 10^{10} \text{ Nm}^2$, and $L = 32.6 \text{ m}$ for each span. The shaking table model is given by Eq. (24), which is the result of the experiments described in Section 4:

$$G_{ST}(s) = e^{-0.006s} \frac{7396s + 1.656 \times 10^6}{s^3 + 134.1s^2 + 3.037 \times 10^4s + 1.649 \times 10^6} \quad (24)$$

The stability of the real-time hybrid testing system is influenced by various factors, including the frequency, damping, mass of the substructure, and the dynamic attributes of the loading system. To quantitatively assess the system's stability, the concept of stability margin is employed. Within current stability studies of real-time hybrid tests, the stability margin is usually described as a multiple of the mass ratio between physical and numerical substructures at critical stability in comparison to the mass ratio in the actual experiments [25,26]. Considering the RHT at a vehicle speed of 300 km/h as an example, Fig. 6 shows the change in the spectral radius in Eq. (23) as the vehicle-bridge mass ratio increases. According to the prediction results, the system becomes unstable when the vehicle-bridge mass ratio exceeds 2.44.

To verify the above predictions, vehicle parameters were taken at vehicle-bridge mass ratios of 2.498 and 2.323, respectively. The time history of the vehicle-bridge coupled RHT was simulated using Eq. (15). The input F_g was considered the pulse function, and x_s was set to zero so that the stability could be judged by whether the displacement time history decayed. The bridge displacements at the wheel-rail contact point are shown in Fig. 7(a). In addition, the spectral radius of the multiplied state matrices, representing the transition matrix between the states at i -th time and initial times, is plotted in Fig. 7(b). It can be seen that the simulated stability of the time-varying RHT matches the prediction based on the proposed stability criterion. When the vehicle-bridge mass ratio was 2.498, the numerical substructure displacement of the system in free vibration exhibited a divergent trend. By contrast, when the vehicle-bridge mass ratio was 2.323, the displacement exhibited an attenuation trend. Although the spectral radius of the cumulative state matrix sometimes exceeds 1 when the mass ratio is 2.323, since the bridge state is updated to zero every time the train reaches a new span, the spectral radius decreases sharply, eventually making the entire system stable. Additionally, the spectrum radius of the cumulative state matrix at each bridge span's beginning changed exponentially, as shown in the green and blue squares in Fig. 7. Therefore, the spectral radius of the product of all state matrices during the vehicle traveling on one single bridge span can be used as the stability criterion.

Table 1
Parameters of the quarter-car model.

Parameters	m_c/kg	$k_c/(\text{kN/m})$	$c_c(\text{kNs/m})$	m_b/kg	$k_b(\text{kN/m})$	$c_b(\text{kNs/m})$	$m_w(\text{kg})$
Value	33,786	900	40	4112	7088	80	5068

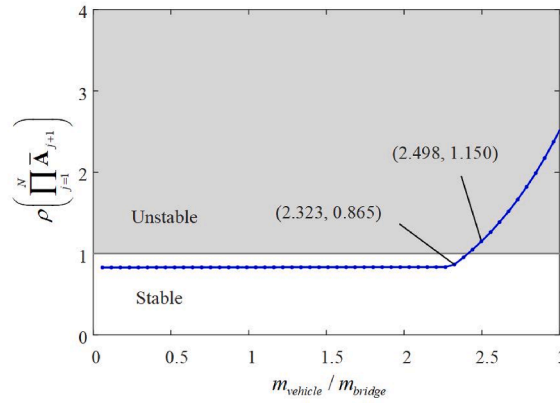


Fig. 6. Spectral radius in the proposed stability criterion with increasing vehicle-bridge mass ratios at a vehicle speed of 300 km/h.

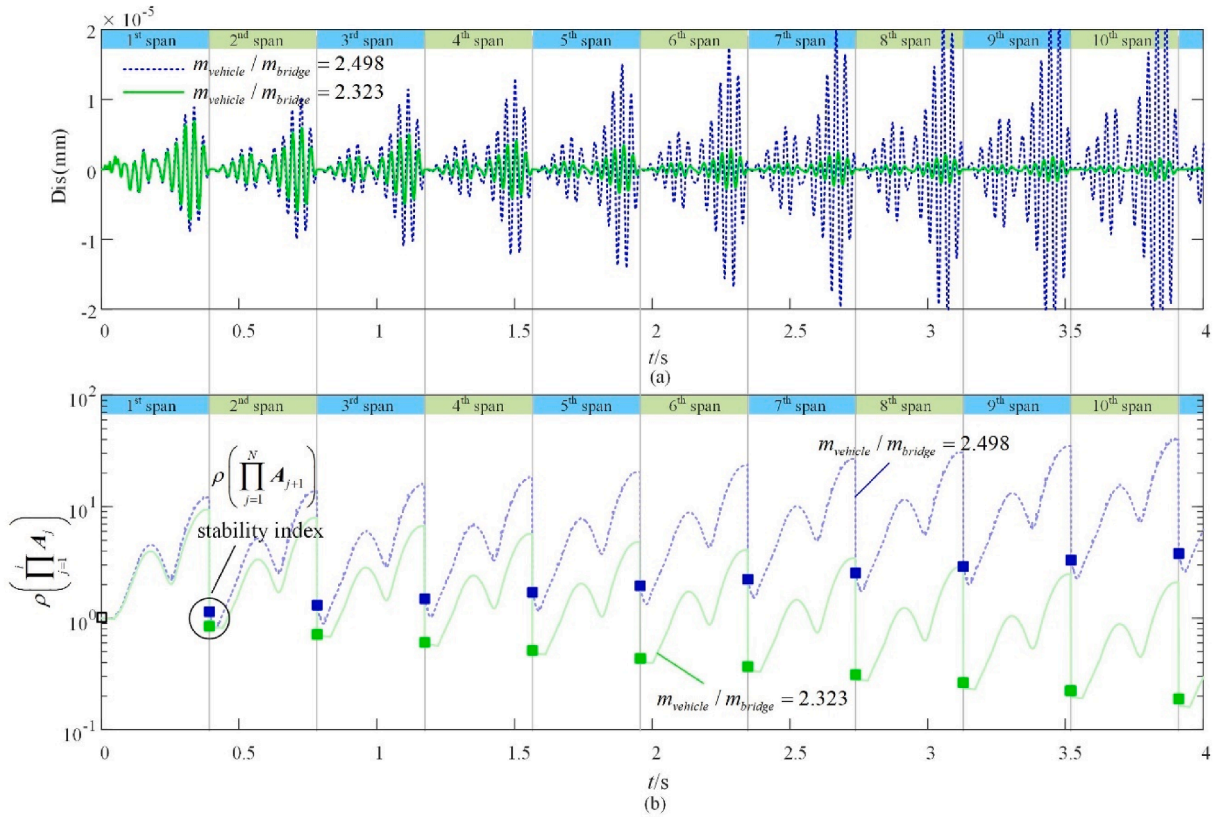


Fig. 7. Characteristics of manifest stability and instability: (a) Time history of numerical substructure displacement, (b) spectral radius of the cumulative state matrix with vehicle-bridge mass ratios of 2.498 and 2.323, respectively.

3.3. Prediction of stability margin

A closed-loop feedback system can be either stable or unstable. This stable/unstable characteristic is called absolute stability, which can be determined using the above stability criterion. However, stability prediction assumes that the RHTT is a known linear system, whereas actual experiments inevitably have nonlinearity or uncertainty. Therefore, predicting the relative stability margin of the RHTT is more meaningful. Referring to the concept of the gain margin in the existing research [25,26,38], this study adopted the ratio h , defined as the multiple of the vehicle-bridge mass ratio at critical stability in comparison to the mass ratio of the original system, to quantitatively describe the stability margin of the time-varying RHTT.

As shown in Fig. 6, the spectral radius in Eq. (23) showed an increasing trend as the vehicle-bridge mass ratio increased, which

inspired the use of the dichotomy method to obtain the vehicle-bridge mass ratio at critical stability. First, to reduce the solution interval of the dichotomy method to a boundary of (0, 1), the system mass ratio, represented by β , is defined as the ratio of the vehicle mass to the total mass of the coupled system, as shown in Eq. (25). M_N and M_P respectively refer to the total mass of the simply supported beam and the vehicle. The system mass ratio of the original coupled system is denoted as β_0 , and the critical stable system mass ratio is denoted as β_{Cri} . Subsequently, the stability margin can be calculated from β_0 and β_{Cri} , as shown in Eq. (26).

$$\beta = \frac{M_P}{M_P + M_N} \in (0, 1) \quad (25)$$

$$h = \frac{\beta_{Cri}(1 - \beta_0)}{(1 - \beta_{Cri})\beta_0}, \quad h(\text{dB}) = 20\lg(h) \quad (26)$$

The intersection point of the proposed time-invariant spectral radius with 1 in the interval $\rho \in (0, 1)$ can be obtained, where the abscissa correspond to the critical stable system mass ratio. The detailed calculation process for the stability margin is as follows.

- (1). We considered 0 and 1 as the upper and lower boundaries, respectively.
- (2). The mean value of the upper and lower boundaries was considered as the system mass ratio. The vehicle parameters were adjusted to the system mass ratio and substituted into the stability criterion in Eq. (23) to determine whether the system is unstable.
- (3). If the system was determined unstable according to the stability criterion, the current system mass ratio was regarded as the upper boundary. Otherwise, the current mass ratio was considered as the lower boundary if the system was stable.
- (4). Steps (2) and (3) were repeated until the difference between the upper and lower boundaries was less than the specified error to obtain a critically stable system mass ratio. Then the stability margin was calculated by substituting the critically stable system mass ratio into Eq. (26).

The stability margin of the vehicle-bridge coupled RHTT was predicted using the above method, which is referred to as the predicted result. In addition, a series of time-history simulations of the vehicle-bridge coupled RHTT under pulse excitation were performed based on Eq. (15). If the simulated time history shows a divergence trend, the system is judged to be unstable; otherwise, if the time history shows a convergence trend, it is judged to be stable. The critical stability of the simulation was also obtained using the dichotomy method, with the corresponding stability margin regarded as the simulation result. Fig. 8 illustrates a comparison between the predicted and simulated results, demonstrating a perfect agreement.

To examine the applicability of the predicted stability depicted in Fig. 8 under various excitations apart from pulse functions, the moving vehicle gravity and track irregularity were employed as the excitation for the vehicle-bridge coupled system. The vehicle gravity takes the parameters of CRH380A prototype. Track irregularity takes the displacement computed from the power spectrum density of vertical irregularities in Class 6 railroad in the United States [39]. This parameter is quantified as:

$$S_V(\Omega) = \frac{kA_v\Omega^2}{\Omega^2(\Omega^2 + \Omega_c^2)} \quad (27)$$

where $k = 0.25$, $A_v = 0.0339\text{cm}^2\text{rad/m}$, $\Omega_c = 0.8245\text{ rad/m}$. The wavelength of spatial frequency Ω is 2—50 m, and the spatial step is 0.1.

The system mass ratio was adjusted to 1.05 times and 0.95 times the predicted critical mass ratio, as well as the original parameters. Subsequently, time history simulations were conducted. The acceleration time history of the numerical substructure is shown in Fig. 9. When the system mass ratio is slightly larger than the predicted critical value, the acceleration time history diverged. By contrast, when the system mass ratio was slightly smaller than the predicted critical value, the acceleration time history was bounded.

Comparison to the simulation results obtained using the original vehicle and bridge parameters, simulations near the critical stable state exhibit numerous high-frequency components. For instance, considering a speed of 300 km/h, Fig. 10 depicts the Fourier amplitude spectra of simulation results. The results indicate that as the RHTT system shifts from a stable to an unstable state, the value of the Fourier amplitude spectrum increased. Particularly, the frequency component within the range of 20–40 Hz demonstrated a more pronounced increase compared to other frequency components. The high-frequency components resulted from the instability of

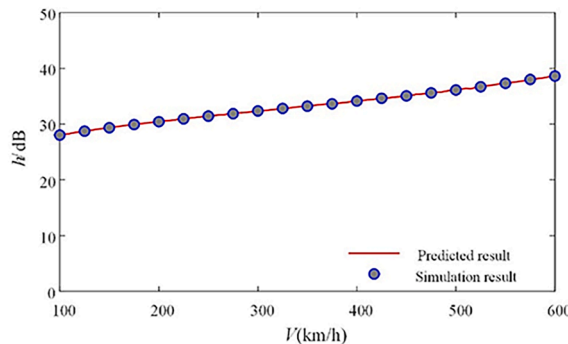


Fig. 8. Predicted stability margins compared to simulated results.

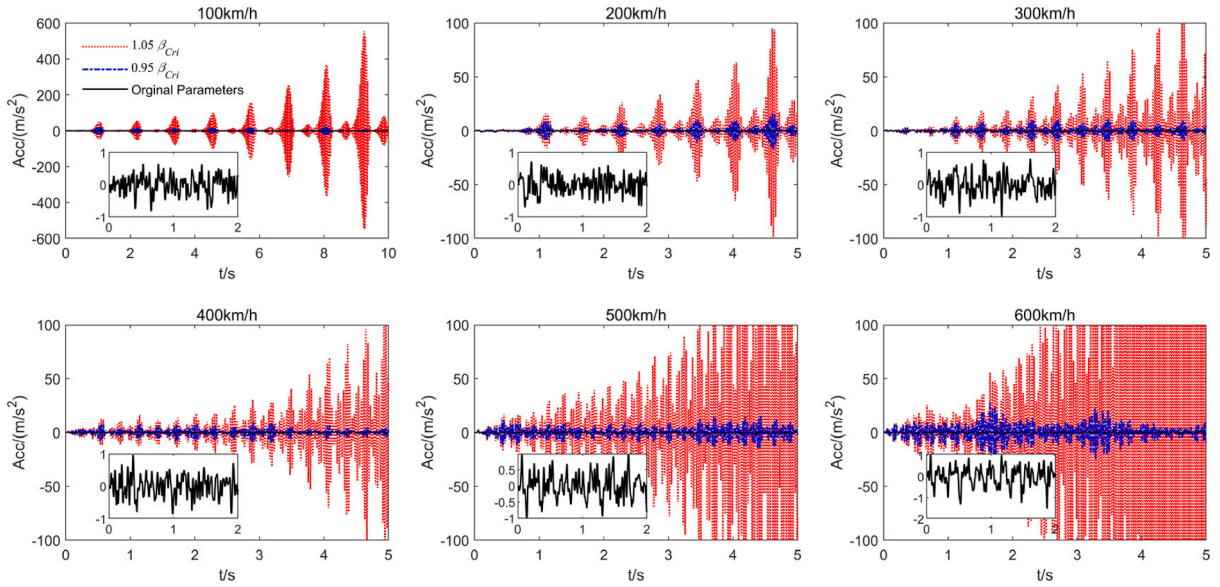


Fig. 9. Acceleration time history of numerical substructure in the manifestly stable and unstable RHTT with the excitation of vehicle moving gravity load.

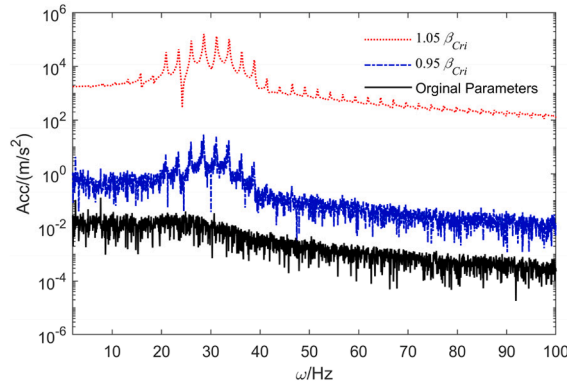


Fig. 10. Fourier spectra of numerical substructure accelerations of vehicle-bridge coupled RHTT systems at 300 km/h with mass ratio of 1.05 and 0.95 critical mass ratio, as well as original vehicle and bridge parameters.

the RHTT system.

4. Experimental verification of stability prediction method for vehicle-bridge coupled RHTT

A series of tests were conducted using an electromagnetic shaking table to validate the accuracy of the developed method in predicting the stability of a practical vehicle-bridge coupled RHTT. The RHTT framework is depicted in Fig. 11. The vehicle is simulated through a two-degree-of-freedom specimen. The numerical substructure is the same as the one employed in Section 3, featuring properties such as $m = 22,691 \text{ kg/m}$, $EI = 5.2 \times 10^{10} \text{ Nm}^2$, and $L = 32.6 \text{ m}$ for each span. Real-time signal processing and transmission were handled using dSPACE. Comprehensive details regarding the physical specimen, loading system, and test results are provided in the subsequent sections.

4.1. Parameters of the physical specimen

The two-degree-of-freedom specimen is shown in Fig. 12 (a). The specimen comprised two mass blocks with normal bearings installed in the middle and vertically supported by two steel shafts to allow the mass blocks to slide horizontally, simulating the vertical movement of the carriage and bogie. The two mass blocks are designed according to the ratio of 4000:1 of the CRH380A carriage and bogie masses respectively. Since the wheelset remains consistently fixed on the shaking table, a virtual mass block was employed to

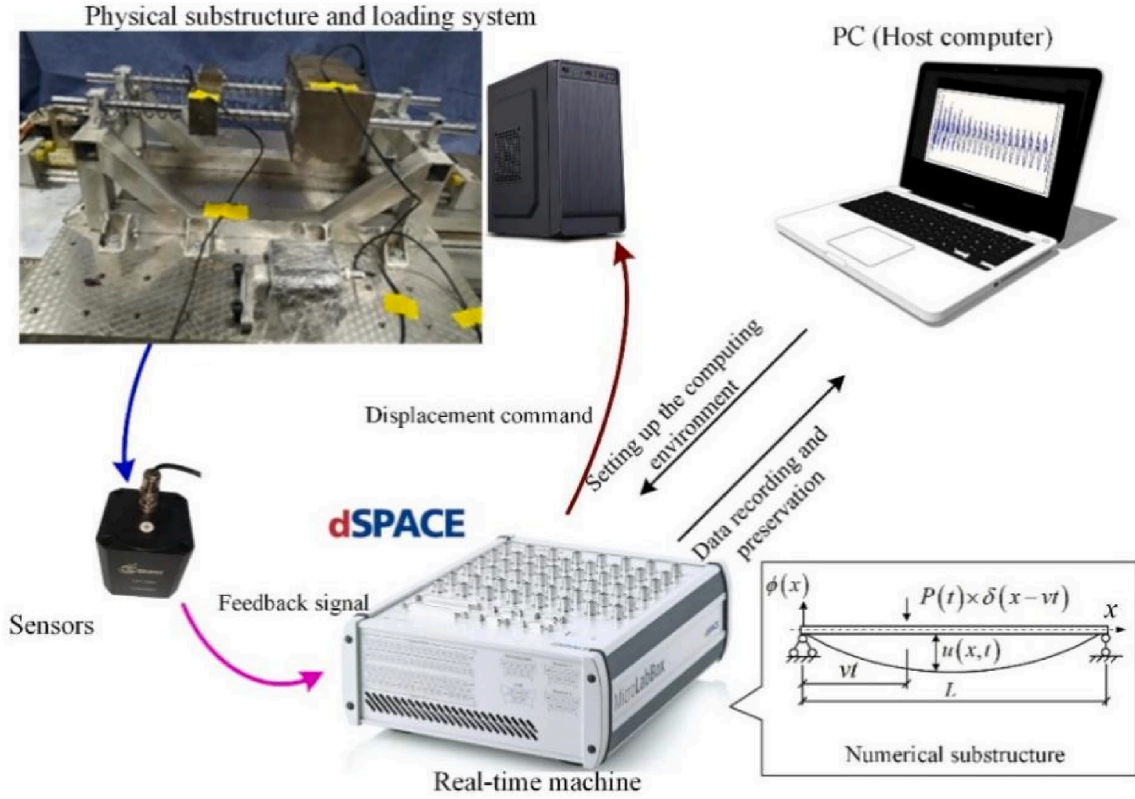


Fig. 11. Schematic of the conducted vehicle-bridge coupled RTHT.

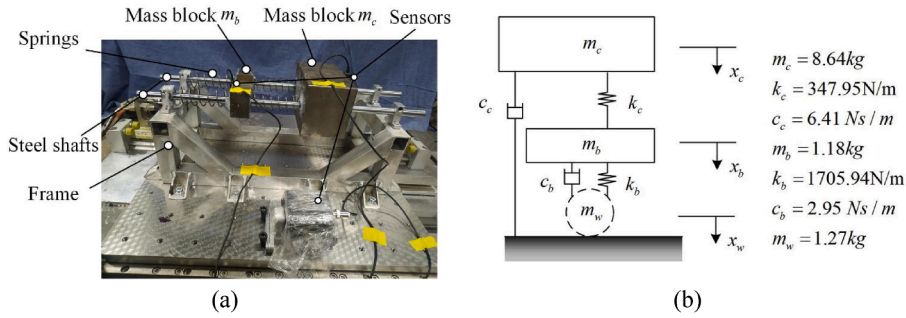


Fig. 12. (a) Picture and (b) dynamic model of the physical specimen.

replicate the wheelset's behavior, as shown by the dashed circle in Fig. 12 (b). The mass of the virtual mass block is set at 1/4000 of the wheelset mass of CRH380A. Simultaneously, the measured acceleration of the shaking table is considered the acceleration attributed to this virtual wheelset. The mass blocks were connected with springs to simulate the primary and secondary suspensions. The primary energy dissipation within the specimen originates from the contact between the bearings within the mass blocks and the steel shafts. This interaction was approximated as viscous damping. Because the two mass blocks are in direct contact with the shaft, the dynamic model should be as shown in Fig. 12 (b), where the damping mechanism differs from the quarter-car model in Fig. 4. However, the purpose of this experiment is to validate the time-varying stability method developed in this paper for assessing the stability of actual experiments. Despite the damping of the specimen differing from the vehicle model depicted in Fig. 4, the RTHT system is still a time-varying system. Therefore, the difference between the specimen and quarter-car models was acceptable. The frequency and damping ratio of the specimen were measured through the independent free vibration of the two mass blocks. Fig. 13 illustrates the comparison between the measured acceleration time history and the identification results of two free vibration mass blocks. Note that when one mass block was in free vibration, the other mass block was fixed. The stiffness and damping coefficients were then calculated, as shown in Fig. 12 (b). The accelerations of the two mass blocks and shaking table were measured using acceleration sensors to calculate the inertial force of the physical substructure, which was then multiplied by the scale ratio to simulate different mass ratio between vehicle

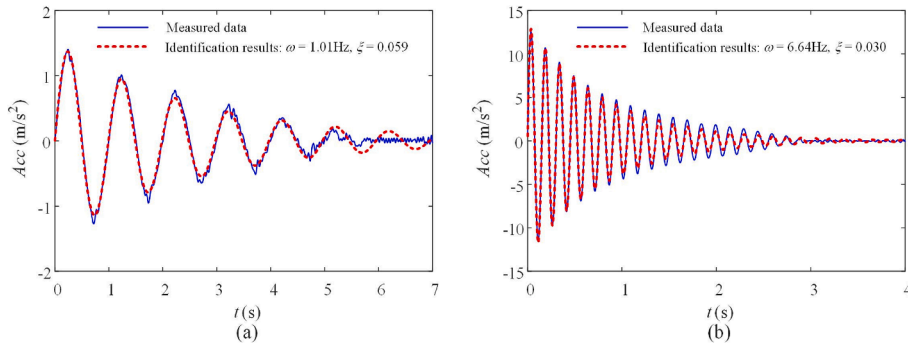


Fig. 13. Comparison between the measured acceleration time history and the identification results of free vibration: (a) carriage mass block and (b) bogie mass block.

and bridge.

4.2. Identification of the shaking table

An electromagnetic shaking table (0.5 m*0.5 m) was used. A sweep signal with a frequency band of 0–50 Hz was input into the shaking table for system identification. The identified shaking table model is expressed in Eq. (24). A comparison between the identified model and measured data is shown in Fig. 14.

4.3. Stability of the RTHT

Given that the stability of the linear system remains unaltered by external excitation [38], and the significant response primarily stems from the instability, to streamline the assessment of the actual RTHT stability, the experiment was simplified by employing the gravitational force of CRH380A as the primary excitation. Furthermore, to prevent potential damage resulting from excessive acceleration in an unstable test, the shaking table ceased loading when the acceleration of the numerical substructure surpassed 40 m/s². The mass ratio of the vehicle to the overall coupled system can be adjusted by changing the feedback gain of the measured reaction force of the physical specimen. A series of tests were conducted for vehicle speeds ranging from 100 to 600 km/h, in which the feedback gain was adjusted to obtain stable and unstable tests. Fig. 15 presents the accelerations of the numerical substructure for both critically unstable and critically stable tests across six scenarios, ranging from speeds of 100 to 600 km/h. The hallmark for differentiating instability from stability lies in the clear divergence trend observed in the acceleration of the unstable test, contrasting with the stable test that, despite sharing a large acceleration amplitude, demonstrates a bounded trend across its overall time history. Given the proximity to critical stability, these selected experiments exhibit larger acceleration amplitudes, accompanied by prominent high-frequency components. In Fig. 16, depicting a test at 300 km/h with $h = 33.2$ dB, the Fourier spectrum of numerical substructure acceleration is presented. The test at critical stability displayed a concentrated frequency component within the 20–40 Hz range. This unstable frequency band closely matches simulation results in Fig. 10, validating the test results.

Moreover, by utilizing the identified parameters of the shaking table and vehicle specimen, the proposed method can be used to predict the stability of the test system, referred to as the predicted results. Fig. 17 illustrates the stability margins corresponding to those critical instability and stability tests, facilitating comparison with predicted stability margins. The alignment between the predicted and the experimental results validates the capability of the proposed method to precisely characterize the relative stability of the vehicle-bridge coupled RTHT system.

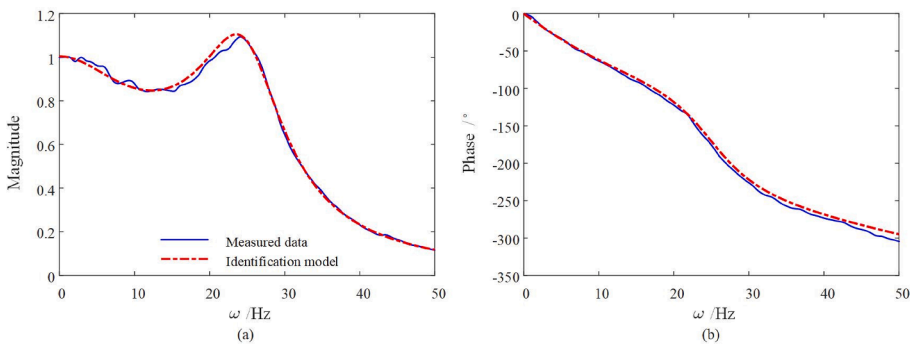


Fig. 14. Modelled and measured frequency response of the shaking table: (a) magnitude, (b) phase.

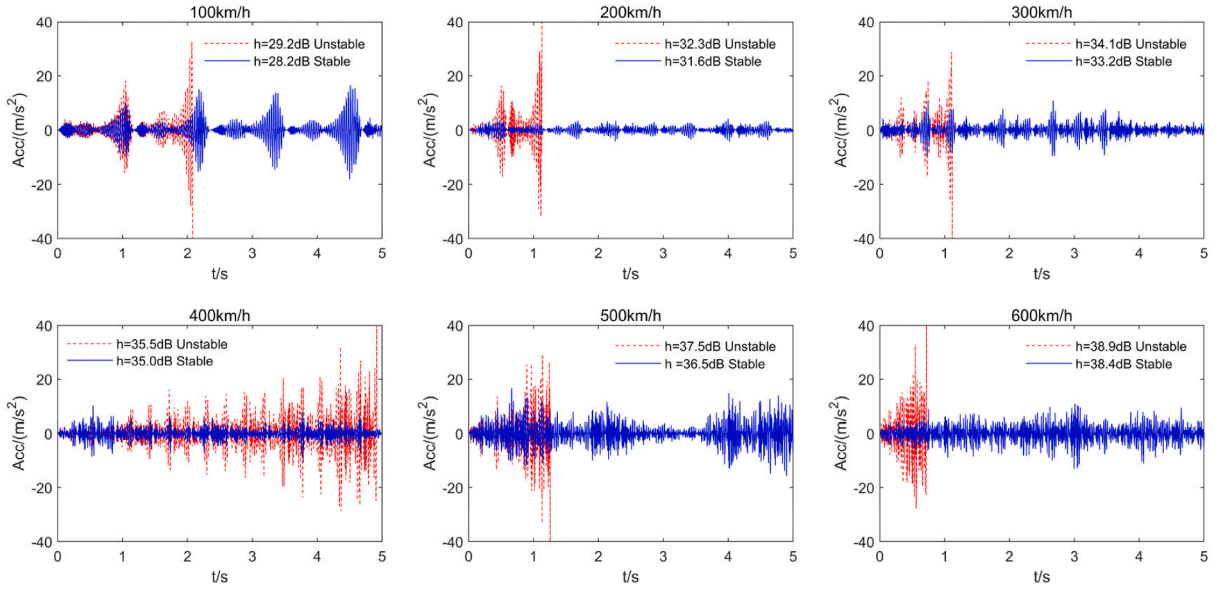


Fig. 15. Time history examples of stable and unstable test.

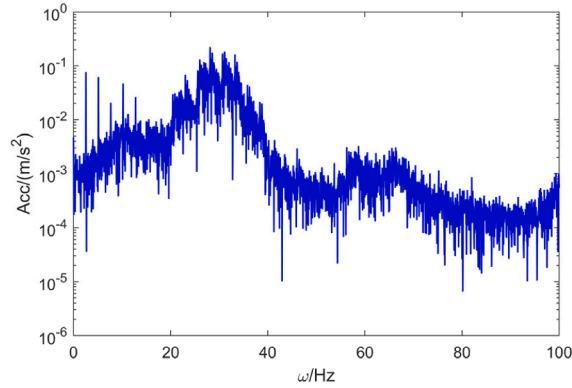


Fig. 16. Fourier spectrum of numerical substructure acceleration test result at critically stable state with vehicle speed of 300 km/h.

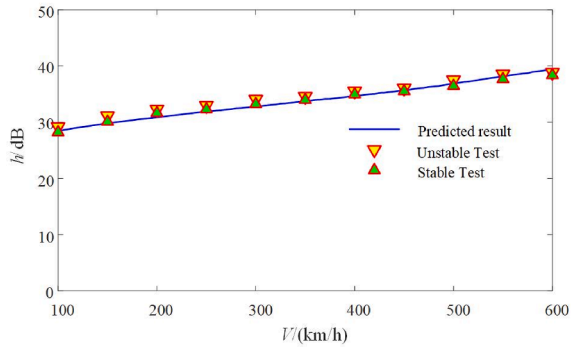


Fig. 17. Comparison of the stability margin of the vehicle-bridge RHT obtained based on practical tests and the proposed stability prediction methods.

5. Parametric study on the stability and accuracy of the vehicle-bridge coupled RHTT

Stability and accuracy are the critical properties of real-time hybrid testing. The error in the RHTT mainly originates from the integral-solving error of the numerical substructure and the loading error in the physical test system, including the physical substructure and shaking table. Existing research shows that when the discrete time step of the numerical integration is sufficiently small (such as 1 ms), the solving error of the numerical substructure can be ignored [26]. In addition, the parameters of trains and bridges are basically known and fixed. Therefore, this study only focuses on the influence of the shaking table performance on the accuracy and stability of the RHTT. The shaking table was modelled as a transfer function (owing to the closed-loop actuator and proportional controller) with a pure time delay (owing to discrete time elements in the controller and filters). To simulate the different performances of the shaking table, the time delay τ was taken as 1, 3, 5, 7, and 10 ms, respectively, in the following simulation, and the transfer functions of the shaking table, G_{ts} , were taken as G_1 , G_2 , and G_3 respectively, as shown in Eq. (28) and Fig. 16. G_2 is the transfer function identified from the shaking table used in this study. Fig. 18 shows that it can accurately load commands within ~ 30 Hz; hence, the effective frequency of G_2 is 30 Hz. G_1 is a transfer function identified from a 3 m * 3 m hydraulic shaking table, and its effective frequency is ~ 10 Hz. G_3 is a transfer function identified from another 0.5 m * 0.5 m electromagnetic shaking table [40], which performance is better than that of the shaking table used in this study, and the effective frequency is ~ 45 Hz.

$$\begin{aligned} G_1(s) &= \frac{7.53 \times 10^7}{s^4 + 406.90s^3 + 3.75 \times 10^4 s^2 + 2.21 \times 10^6 s + 7.30 \times 10^7} \\ G_2(s) &= \frac{7396s + 1.656 \times 10^6}{s^3 + 134.1s^2 + 3.037 \times 10^4 s + 1.649 \times 10^6} \\ G_3(s) &= \frac{8.51 \times 10^6}{s^3 + 234.60s^2 + 8.88 \times 10^4 s + 8.50 \times 10^6} \end{aligned} \quad (28)$$

The stability of the RHTT is represented by a defined stability margin h . The accuracy was quantified using the displacement similarity rate S_d and acceleration similarity rate S_a , as defined in Eq. (29).

$$\begin{cases} S_d = \left(1 + \frac{\sum (x_e(t) - x_g(t))^2}{\sum x_e(t)^2} \right)^{-1} \times 100\% \\ S_a = \left(1 + \frac{\sum (\ddot{x}_e(t) - \ddot{x}_g(t))^2}{\sum \ddot{x}_e(t)^2} \right)^{-1} \times 100\% \end{cases}, \quad (29)$$

where x_e and $\ddot{x}_e$ refer to the displacement and acceleration of the bridge in the simulated vehicle-bridge coupled system, respectively, and x_g and $\ddot{x}_g$ refer to the displacement and acceleration of the shaking table in the RHTT simulation, respectively. A similarity rate of 100 % indicated perfect accuracy of the RHTT, whereas a lower value indicated poorer accuracy.

The proposed method was used to predict the stability margins of RHTTs. The results are plotted in Fig. 19, with each subfigure showing the stability margins for different shaking-table time delays and vehicle speeds. It can be observed that the stability margins increase at higher vehicle speeds. Larger shaking table time delays decreased the stability margins, aligning with the trends observed in linear time-invariant RHTT systems. This indicates that the time delay of the shaking table should be maximally reduced to stabilize the RHTT. For RHTTs with large time delays ($\tau > 5$ ms), the stability decreased and then increased as the effective frequency of the shaking table increased. For RHTTs with a small time delay ($\tau < 5$ ms), the stability declined overall as the effective frequency increased. In general, an increase in the effective frequency of the shaking table can reduce the stability of the RHTT. In addition, Fig. 19 shows that the real-time hybrid test has sufficient stability margins for the vehicle-bridge coupled system composed of a

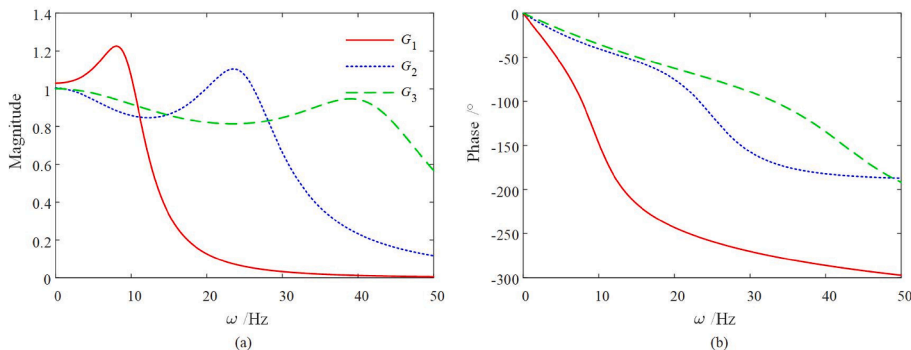


Fig. 18. Frequency response of the transfer functions G_1 , G_2 , and G_3 : (a) magnitude, (b) phase.

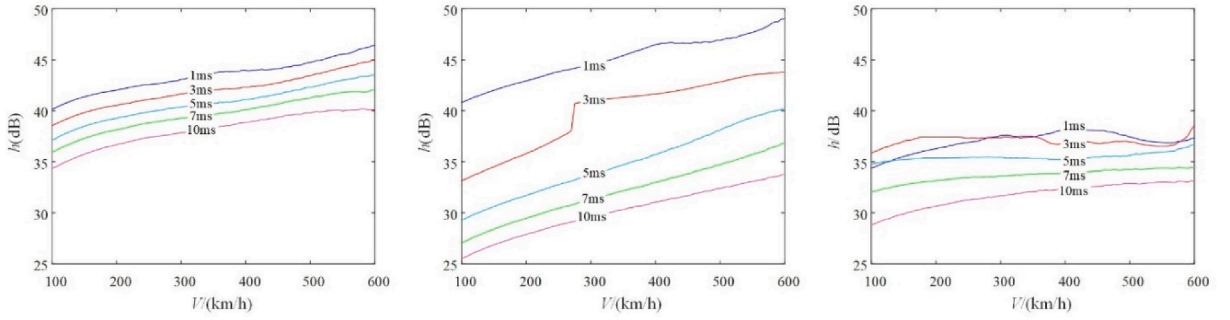


Fig. 19. Stability margin of the vehicle-bridge coupled RHTT with different transfer functions of the shaking table: (a) G_1 , (b) G_2 , and (c) G_3 .

quarter-car model and simply supported beams. The minimum stability margin exceeds 25 dB, which is much higher than the recommended 6 dB from existing studies [38]. However, in future optimizations of high-speed railway bridges, potential changes in the cross-sectional parameters of the bridge might occur. It remains uncertain whether the vehicle-bridge coupled RHTT system will remain stable under such modified conditions. The method proposed in this paper offers a valuable reference for predicting the test stability amid these changes.

The accuracy of the vehicle-bridge coupled real-time hybrid testing (RHTT) system was analyzed through time-history simulations conducted with varied time delays and transfer functions. As shown in Fig. 20, the displacement and acceleration similarity rates increased as the time delay decreased and the effective frequency of the shaking table increased. Reducing the time delay provided some accuracy gains, but increasing the effective frequency proved substantially more impactful. However, the similarity rate improvements plateaued once the effective frequency surpassed 30 Hz. These results highlight that sufficiently high shaking table effective frequencies above 30 Hz are critical to achieve robust accuracy in RHTT systems.

As the time delay increased, both accuracy and stability declined, highlighting the importance of minimizing time delays. However, increasing the effective frequency of the shaking table divergently impacted the accuracy and stability; the accuracy improved while the stability decreased. Therefore, for a practical vehicle-bridge coupled RHTT, the effective loading frequency of the shaking table should be increased while ensuring a sufficient stability margin.

6. Conclusions

Conventional real-time hybrid testing (RHTT) assumes a fixed interface between the numerical and physical substructures.

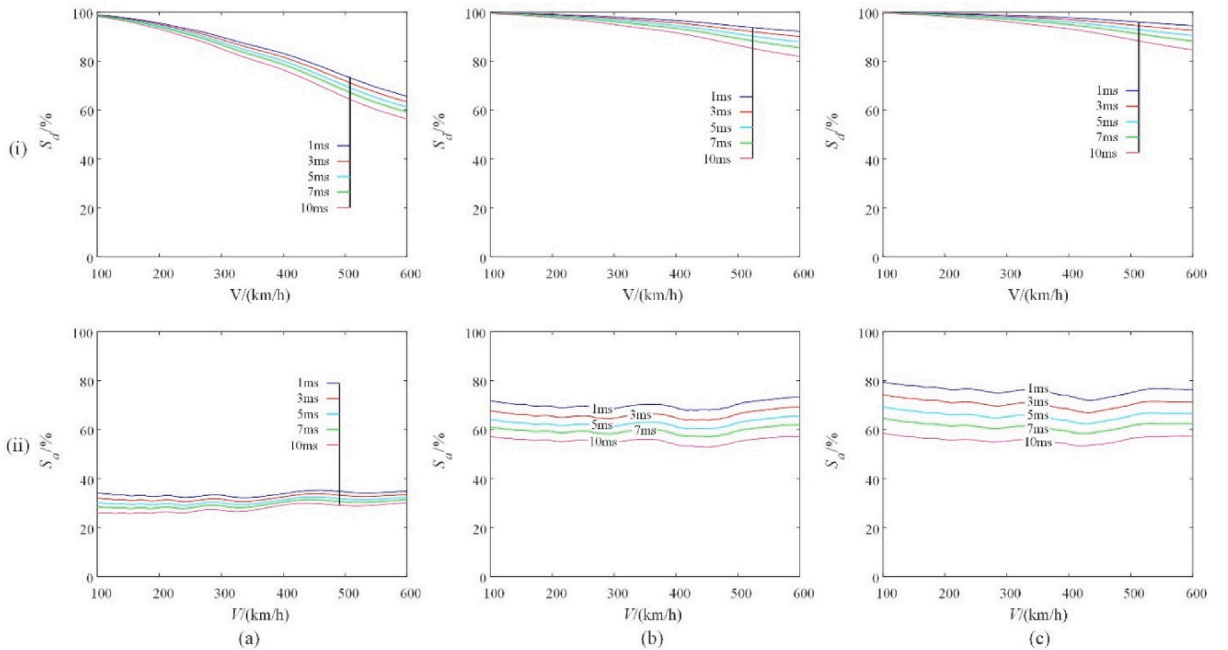


Fig. 20. (a) Displacement similarity rate and (b) acceleration similarity rate of the vehicle-bridge coupled RHTT with the transfer functions of (a) G_1 , (b) G_2 , and (c) G_3 of the shaking table.

However, the application of RTHT to vehicle-bridge coupled systems introduces a time-varying interface, which poses challenges for stability analysis. To address this issue, this paper introduces a novel stability prediction method tailored for RTHT systems with time variability. The method was used to analyze the stability and accuracy of an RTHT framework including a quarter-car model coupled with simply supported bridge beams. The conclusions are as follows.

1. A dynamic model for time-varying vehicle-bridge coupled RTHT was established based on a discrete state-space representation. This model comprehensively integrates the time-varying vehicle-bridge interaction and shaking table dynamics, providing a foundation for the development of a stability prediction method tailored to time-varying RTHT systems
2. The time-invariant characteristic of the time-varying system was derived from the periodic time variability of the vehicle-bridge coupled system at a constant vehicle speed. Based on this characteristic, an innovative method that can predict the stability margin of the time-varying RTHT was further proposed. Experiments verified the accuracy of the proposed stability prediction method
3. The time delay in the shaking table is detrimental to both the accuracy and stability of the vehicle-bridge coupled RTHT, thus it should be minimized as much as possible in practical experiments. Increasing the effective loading frequency of the shaking table can improve the accuracy but may reduce the stability. Therefore, to enhance the accuracy of practical vehicle-bridge coupled RTHT, the effective loading frequency of the shaking table should be increased while ensuring that the RTHT retains an adequate stability margin

This paper primarily tackles the challenges posed by time variability in predicting RTHT stability. Yet, beyond time variability, the stability prediction of the vehicle-bridge coupled RTHT system faces other complexities, including time asynchrony and multi-interface interactions. Consequently, further development of the proposed method is essential to comprehensively address the influence of all parameters on stability in future endeavors.

CRedit authorship contribution statement

Hao Liu: Writing – original draft, Data curation. **Zhenyun Tang:** Project administration, Methodology. **Ryuta Enokida:** Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix I. Expressions of the parameters in the numerical substructure

The modal mass matrix $\mathbf{M}_q$, modal damping matrix $\mathbf{C}_q$, modal stiffness matrix $\mathbf{K}_q$, and modal shape matrix $\boldsymbol{\phi}$ in Eq. (2) are expressed as Eq. (28):

$$\boldsymbol{\phi}(x) = \begin{Bmatrix} \sin\left(\frac{\pi x}{L}\right) \\ \sin\left(\frac{2\pi x}{L}\right) \\ \vdots \\ \sin\left(\frac{n_q \pi x}{L}\right) \end{Bmatrix}, \mathbf{M}_q = \begin{bmatrix} m \int_0^L \phi_1(x)^2 dx & 0 & \cdots & 0 \\ 0 & m \int_0^L \phi_2(x)^2 dx & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & m \int_0^L \phi_{n_q}(x)^2 dx \end{bmatrix} \quad (28)$$

$$\mathbf{C}_q = \mathbf{M}_q \begin{bmatrix} 2\xi_1 \omega_1 & 0 & \cdots & 0 \\ 0 & 2\xi_2 \omega_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & 2\xi_{n_q} \omega_{n_q} \end{bmatrix}, \mathbf{K}_q = \mathbf{M}_q \begin{bmatrix} \omega_1^2 & 0 & \cdots & 0 \\ 0 & \omega_2^2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \omega_{n_q}^2 \end{bmatrix}$$

The expressions for the state matrix $\mathbf{A}_b$, input matrix $\mathbf{B}_{b,i}$, and output matrix $\mathbf{C}_{b,i}$ of the numerical substructure in Eq. (3) and Eq. (6) are expressed as Eq. (29).

$$\mathbf{A}_b = \begin{bmatrix} \hat{\mathbf{K}}^{-1}(a_0 \mathbf{M}_q + a_1 \mathbf{C}_q) & \hat{\mathbf{K}}^{-1}(a_2 \mathbf{M}_q + a_4 \mathbf{C}_q) & \hat{\mathbf{K}}^{-1}(a_3 \mathbf{M}_q + a_5 \mathbf{C}_q) \\ \hat{\mathbf{K}}^{-1} a_0 a_7 (a_0 \mathbf{M}_q + a_1 \mathbf{C}_q) - a_0 a_7 \mathbf{E} & \hat{\mathbf{K}}^{-1} a_0 a_7 (a_2 \mathbf{M}_q + a_4 \mathbf{C}_q) - (a_2 a_7 - 1) \mathbf{E} & \hat{\mathbf{K}}^{-1} a_0 a_7 (a_3 \mathbf{M}_q + a_5 \mathbf{C}_q) - (a_3 a_7 - a_6) \mathbf{E} \\ \hat{\mathbf{K}}^{-1} a_0 (a_0 \mathbf{M}_q + a_1 \mathbf{C}_q) - a_0 \mathbf{E} & \hat{\mathbf{K}}^{-1} a_0 (a_2 \mathbf{M}_q + a_4 \mathbf{C}_q) - a_2 \mathbf{E} & \hat{\mathbf{K}}^{-1} a_0 (a_3 \mathbf{M}_q + a_5 \mathbf{C}_q) - a_3 \mathbf{E} \end{bmatrix}, \quad (29)$$

$$\mathbf{B}_{b,i} = \begin{Bmatrix} \hat{\mathbf{K}}^{-1} \\ \hat{\mathbf{K}}^{-1} a_0 a_7 \\ -\hat{\mathbf{K}}^{-1} a_0 \mathbf{M}_q \end{Bmatrix} \times \boldsymbol{\phi}(vt_i), \mathbf{C}_{b,i} = \left\{ \boldsymbol{\phi}^T(vt_i) \quad \mathbf{0}_{1 \times 2n_q} \right\}, \mathbf{z}_{b,i} = \begin{Bmatrix} \mathbf{q}_i \\ \dot{\mathbf{q}}_i \\ \ddot{\mathbf{q}}_i \end{Bmatrix},$$

$$\hat{\mathbf{K}} = \mathbf{K}_q + a_0 \mathbf{M}_q + a_1 \mathbf{C}_q$$

where a_0 – a_7 are coefficients in the Newmark- β method, as shown in Eq. (30):

$$a_0 = \frac{1}{\beta \Delta t^2}, a_1 = \frac{\gamma}{\beta \Delta t^2}, a_2 = \frac{1}{\beta \Delta t}, a_3 = \frac{1}{2\beta} - 1, \quad (30)$$

$$a_4 = \frac{\gamma}{\beta} - 1, a_5 = \frac{\Delta t}{2} \left(\frac{\gamma}{\beta} - 2 \right), a_6 = \Delta t(1 - \gamma), a_7 = \gamma \Delta t$$

Appendix II. . Expressions of the parameters in the physical test system

The matrices in the discrete state–space representation of the pure delay term (Eq. 12(a)) in Eq. (13) are:

$$\mathbf{z}_{t,i} = \left\{ \ddot{x}_{c,i-n_d} \quad \ddot{x}_{c,i+1-n_d} \quad \cdots \quad \ddot{x}_{c,i} \right\}^T, \mathbf{A}_d = \begin{bmatrix} \mathbf{0}_{n_d \times 1} & \mathbf{E}_{n_d} \\ \mathbf{0} & \mathbf{0}_{1 \times n_d} \end{bmatrix}, \mathbf{B}_d = \begin{Bmatrix} \mathbf{0}_{n_d \times 1} \\ 1 \end{Bmatrix}, \mathbf{C}_d = \{ 1 \quad \mathbf{0}_{1 \times n_d} \} \quad (31)$$

where n_d refers that $\tau = n_d \times \Delta t$. $\mathbf{0}$ is a matrix in which all elements are 0, and the subscripts represent the number of rows and columns in turn. $\mathbf{E}$ is the identity matrix, and the subscript indicates its dimensions.

The transfer function term of the shaking-table model (Eq. 12(b)) can be decomposed into multiple single-order transfer functions as follows:

$$F_f(s) = x_{g0}(s) G_{st}(s) G_v(s) = x_{g0}(s) \sum_{i=1}^{n_c} \left(\frac{f_i}{s - \lambda_i} + k_0 \right) \quad (32)$$

where n_c is the order of the transfer-function term. Eq. (32) can be transformed into a continuous state-space representation, as shown in Eq. (33):

$$\begin{cases} \dot{\mathbf{z}}_c(t) = \lambda_c \mathbf{z}_c(t) + \mathbf{I}_c \mathbf{x}_{g0}(t) \\ F_f(t) = \mathbf{C}_c \mathbf{z}_c(t) + \mathbf{D}_c \mathbf{x}_{g0}(t) \end{cases}$$

$$\lambda_c = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & \\ & & \ddots \\ 0 & 0 & \lambda_{n_c} \end{bmatrix}, \mathbf{I}_c = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \mathbf{C}_c = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_{n_c} \end{bmatrix}^T, \mathbf{D}_c = k_0$$
(33)

The fourth order Runge–Kutta method shown in equation (34) was used to solve Eq. (33).

$$\begin{cases} \mathbf{z}_{c,i+1} = \mathbf{z}_{c,i} + \frac{1}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \\ F_{fi+1} = \mathbf{C}_c \mathbf{z}_{c,i+1} + \mathbf{D}_c \mathbf{x}_{g0,i+1} \\ \mathbf{k}_1 = \Delta t \times (\lambda_c \mathbf{z}_{c,i} + \mathbf{I}_c \mathbf{x}_{g0,i}) \\ \mathbf{k}_2 = \Delta t \times \left[\lambda_c \left(\mathbf{z}_{c,i} + \frac{\mathbf{k}_1}{2} \right) + \mathbf{I}_c \mathbf{x}_{g0,i} \right] \\ \mathbf{k}_3 = \Delta t \times \left[\lambda_c \left(\mathbf{z}_{c,i} + \frac{\mathbf{k}_2}{2} \right) + \mathbf{I}_c \mathbf{x}_{g0,i} \right] \\ \mathbf{k}_4 = \Delta t \times [\lambda_c (\mathbf{z}_{c,i} + \mathbf{k}_3) + \mathbf{I}_c \mathbf{x}_{g0,i}] \end{cases}$$
(34)

The above process can be organised into a state–space representation. The expressions for state matrix $\mathbf{A}_c$, input matrices $\mathbf{B}_c$ can be expressed as Eq. (35).

$$\begin{aligned} \mathbf{A}_c &= \left(\mathbf{E} + \frac{\Delta t}{6}(\lambda_1 + 2\lambda_2 + 2\lambda_3 + \lambda_4) \right), \\ \mathbf{B}_c &= \frac{\Delta t}{6}(\bar{\mathbf{I}}_1 + 2\bar{\mathbf{I}}_2 + 2\bar{\mathbf{I}}_3 + \bar{\mathbf{I}}_4), \end{aligned}$$
(35)

where the expressions for λ_1 – λ_4 , and $\bar{\mathbf{I}}_1$ – $\bar{\mathbf{I}}_4$ are:

$$\begin{aligned} \lambda_1 &= \lambda_c, \lambda_2 = \lambda_1 + \frac{\Delta t \times \lambda_1^2}{2}, \lambda_3 = \lambda_1 + \frac{\Delta t \times \lambda_1 \lambda_2}{2}, \lambda_4 = \lambda_1 + \frac{\Delta t \times \lambda_1 \lambda_3}{2}, \\ \bar{\mathbf{I}}_1 &= \mathbf{I}_c, \bar{\mathbf{I}}_2 = \frac{\Delta t \times \lambda_1 \bar{\mathbf{I}}_1 + 3\bar{\mathbf{I}}_1}{2}, \bar{\mathbf{I}}_3 = \Delta t \times \lambda_1 \bar{\mathbf{I}}_2 + \bar{\mathbf{I}}_1, \bar{\mathbf{I}}_4 = \Delta t \times \lambda_1 \bar{\mathbf{I}}_3 + \bar{\mathbf{I}}_1 \end{aligned}$$
(36)

Appendix III. . Expressions of the RTHT on the train–bridge coupled system

The expressions for state vector $\mathbf{Z}_i$, state matrix $\mathbf{A}_i$, and input matrix $\mathbf{B}_i$ in Eq. (15) are expressed as Eq. (37):

$$\begin{aligned} \mathbf{A}_{i+1} &= \begin{pmatrix} \mathbf{E}_{n_b+n_d+n_c} - \begin{bmatrix} \mathbf{0}_{n_b \times n_b} & \mathbf{0}_{n_b \times 1} & \mathbf{0}_{n_b \times n_d} & \mathbf{0}_{n_b \times 1} & \mathbf{0}_{n_b \times n_c} & \mathbf{B}_{b,i+1} \\ \mathbf{C}_{b,i+1} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_d} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_c} & \mathbf{0}_{1 \times 1} \\ \mathbf{0}_{n_d \times n_b} & \mathbf{B}_d & \mathbf{0}_{n_d \times n_d} & \mathbf{0}_{n_d \times 1} & \mathbf{0}_{n_d \times n_c} & \mathbf{0}_{n_d \times 1} \\ \mathbf{0}_{1 \times n_b} & \mathbf{0}_{1 \times 1} & \mathbf{C}_d & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_c} & \mathbf{0}_{1 \times 1} \\ \mathbf{0}_{n_c \times n_b} & \mathbf{0}_{n_c \times 1} & \mathbf{0}_{n_c \times n_d} & \mathbf{0}_{n_c \times 1} & \mathbf{0}_{n_c \times n_c} & \mathbf{0}_{n_c \times 1} \\ \mathbf{0}_{1 \times n_b} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_d} & \mathbf{D}_c & \mathbf{C}_c & \mathbf{0}_{1 \times 1} \end{bmatrix} \\ \begin{bmatrix} \mathbf{A}_b & \mathbf{0}_{n_b \times 1} & \mathbf{0}_{n_b \times n_d} & \mathbf{0}_{n_b \times 1} & \mathbf{0}_{n_b \times n_c} & \mathbf{0}_{n_b \times 1} \\ \mathbf{0}_{1 \times n_b} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_d} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_c} & \mathbf{0}_{1 \times 1} \\ \mathbf{0}_{n_d \times n_b} & \mathbf{0}_{n_d \times 1} & \mathbf{A}_d & \mathbf{0}_{n_d \times 1} & \mathbf{0}_{n_d \times n_c} & \mathbf{0}_{n_d \times 1} \\ \mathbf{0}_{1 \times n_b} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_d} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_c} & \mathbf{0}_{1 \times 1} \\ \mathbf{0}_{n_c \times n_b} & \mathbf{0}_{n_c \times 1} & \mathbf{0}_{n_c \times n_d} & \mathbf{B}_c & \mathbf{A}_c & \mathbf{0}_{n_c \times 1} \\ \mathbf{0}_{1 \times n_b} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_d} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_c} & \mathbf{0}_{1 \times 1} \end{bmatrix} \end{pmatrix}, \\ \mathbf{B}_{i+1} &= \begin{pmatrix} \mathbf{E}_{n_b+n_d+n_c} - \begin{bmatrix} \mathbf{0}_{n_b \times n_b} & \mathbf{0}_{n_b \times 1} & \mathbf{0}_{n_b \times n_d} & \mathbf{0}_{n_b \times 1} & \mathbf{0}_{n_b \times n_c} & \mathbf{B}_{b,i+1} \\ \mathbf{C}_{b,i+1} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_d} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_c} & \mathbf{0}_{1 \times 1} \\ \mathbf{0}_{n_d \times n_b} & \mathbf{B}_d & \mathbf{0}_{n_d \times n_d} & \mathbf{0}_{n_d \times 1} & \mathbf{0}_{n_d \times n_c} & \mathbf{0}_{n_d \times 1} \\ \mathbf{0}_{1 \times n_b} & \mathbf{0}_{1 \times 1} & \mathbf{C}_d & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_c} & \mathbf{0}_{1 \times 1} \\ \mathbf{0}_{n_c \times n_b} & \mathbf{0}_{n_c \times 1} & \mathbf{0}_{n_c \times n_d} & \mathbf{0}_{n_c \times 1} & \mathbf{0}_{n_c \times n_c} & \mathbf{0}_{n_c \times 1} \\ \mathbf{0}_{1 \times n_b} & \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times n_d} & \mathbf{D}_c & \mathbf{C}_c & \mathbf{0}_{1 \times 1} \end{bmatrix} \end{pmatrix}^{-1} \begin{bmatrix} \mathbf{B}_{b,i+1} & \mathbf{0}_{n_b \times 1} \\ \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times 1} \\ \mathbf{0}_{n_d \times 1} & \mathbf{B}_d \\ \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times 1} \\ \mathbf{0}_{n_c \times 1} & \mathbf{0}_{n_c \times 1} \\ \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times 1} \end{bmatrix}, \mathbf{Z}_{i+1} = \begin{bmatrix} \mathbf{z}_{b,i+1} \\ \mathbf{x}_{I,i+1} \\ \mathbf{z}_{d,i+1} \\ \mathbf{x}_{g0,i+1} \\ \mathbf{z}_{c,i+1} \\ \mathbf{F}_{f,i+1} \end{bmatrix}, \mathbf{Y}_{i+1} = \begin{bmatrix} \mathbf{F}_g \\ \mathbf{x}_{s,i+1} \end{bmatrix} \end{pmatrix}$$
(37)

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